Foundations of Deep Learning **Fall 2026**

Exercise 3: Optimization

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Problem 1 (Gradient Descent for Least-square Problem):
Consider the problem of solving the linear system

$$\mathbf{X}\mathbf{w} = \mathbf{y}$$

where $\mathbf{X} = (\mathbf{x}_i^\top)_{i=1}^n \in \mathbb{R}^{n\times d}$ is the data matrix, $\mathbf{w} \in \mathbb{R}^d$ the parameter vector of the model, and $\mathbf{y} \in \mathbb{R}^n$ the target. We assume that there exists a unique solution provided by the mean-squared loss

$$\mathbf{w}_* = \arg\min_{\mathbf{w}\in\mathbb{R}^d} \left(\frac{1}{2}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 =: \ell(\mathbf{w}) \right).$$

We consider the gradient descent method

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \ell(\mathbf{w}_t) \tag{1}$$

with a fixed step size η.

a) Calculate the gradient $\nabla\ell(\mathbf{w})$ and use it to re-write the gradient descent iterates of Eq. (1).

b) Calculate the Hessian $\nabla^2\ell(\mathbf{w})$ and argue whether or not $\ell(\mathbf{w})$ is convex.

c) Use the following facts.

- Remember that $\mathbf{y} = \mathbf{X}\mathbf{w}_*$;
- Recall the definition of the operator norm of a matrix:

$$\|\mathbf{A}\|_{\text{op}} = \sup\left\{ \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} : \mathbf{x} \in \mathbb{R}^d \text{ with } \mathbf{x} \neq 0 \right\} = \sqrt{\lambda_{\max}(\mathbf{A}^T\mathbf{A})},$$

  where $\lambda_{\max}(\mathbf{A}^T\mathbf{A})$ denotes the largest eigenvalue of $\mathbf{A}^T\mathbf{A}$.
- for any matrix $\mathbf{A} \in \mathbb{R}^{d\times d}$ and vector $\mathbf{v} \in \mathbb{R}^d$, we have $\|\mathbf{A}\mathbf{v}\|_2 \leq \|\mathbf{A}\|_{\text{op}} \cdot \|\mathbf{v}\|_2$, where $\|\mathbf{A}\|_{\text{op}}$ denotes the operator norm of the matrix $\mathbf{A}$;

Show that the following bound on the distance to the optimizer holds

$$\|\mathbf{w}_{t+1} - \mathbf{w}_*\|_2 \leq \|\mathbf{I} - \eta\mathbf{X}^\top\mathbf{X}\|_{\text{op}}^t \cdot \|\mathbf{w}_1 - \mathbf{w}_*\|_2 \tag{2}$$

d) Show that, if $\eta < \frac{1}{\|\mathbf{X}^\top\mathbf{X}\|_{\text{op}}}$, i.e. if $\|\eta\mathbf{X}^\top\mathbf{X}\|_{\text{op}} < 1$, then the Gradient Descent converge, i.e. $\|\mathbf{w}_{t+1} - \mathbf{w}_*\|_2 \to 0$ as $t \to \infty$.

e) One, obviously cheaper, alternative is to only compute the update (indexed by k) based on the gradient of one specific datapoint $\mathbf{x}_i$. This is the updated of *stochastic* gradient descent (SGD), which is arguably the most widely used optimizer in ML:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta\nabla\ell_i(\mathbf{w}_k), \quad i \in \{1,\dots,n\};\ \eta > 0.$$

where we denote $\ell_i(\mathbf{w}) := \frac{n}{2}(\mathbf{x}_i^\top\mathbf{w} - y_i)^2$ for each $i = 1, ..., n$. In each iteration k, a datapoint $\mathbf{x}_i$ is chosen uniformly at random such that $\mathbb{E}[\nabla\ell_i(\mathbf{w}_k)] = \nabla\ell(\mathbf{w}_k)$.

Show that, given a constant positive step size $\eta \leq \frac{1}{2\|\mathbf{X}^\top\mathbf{X}\|_{\text{op}}}$, we have

$$\mathbb{E}\left[\|\mathbf{w}_{k+1} - \mathbf{w}_*\|_2^2\right] \geq \eta^2\mathbb{E}\left[\|\nabla\ell_i(\mathbf{w}_k)\|_2^2\right] > 0$$

In other words, SGD does not converge to the critical point $\mathbf{w}_*$ on average.

Problem 2 (Polyak's momentum on a quadratic objective):
Consider Polyak's momentum method

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t) + \beta(\mathbf{x}_t - \mathbf{x}_{t-1}), \quad \eta > 0, \quad 0 \leq \beta < 1.$$

We study its behavior on a quadratic objective

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{H}\mathbf{x},$$

where $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{H} \in \mathbb{R}^{d \times d}$ is symmetric.

a) Let $\mathbf{H} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ be an eigendecomposition of $\mathbf{H}$, and define

$$\mathbf{z}_t = \mathbf{Q}^\top \mathbf{x}_t.$$

Show that the coordinate z_t^i of $\mathbf{z}_t$ evolve independently.

b) In an eigendirection corresponding to eigenvalue λ_i, show that

$$z_{t+1}^i = (1 + \beta - \eta\lambda_i) z_t^i - \beta z_{t-1}^i.$$

Thus, the behavior of momentum on a quadratic can be understood by studying a scalar second-order recurrence.

c) For the remaining part of the exercise, we will drop the subscript and write z_t instead of z_t^i.
For a linear recurrence with constant coefficients, try a solution of the form $z_t = r^t$. Why is such a form particularly convenient under time shifts?
Derive the characteristic equation

$$r^2 - (1 + \beta - \eta\lambda_i) r + \beta = 0.$$

Explain why convergence in this eigendirection requires both characteristic roots to satisfy $|r| < 1$. Under what condition are the roots complex and what does this mean qualitatively for the optimization trajectory?

d) The loss landscape of a neural network is generally nonconvex. Nevertheless, one can locally study its behavior through Hessian eigendirections. Consider again the characteristic equation

$$r^2 - (1 + \beta - \eta\lambda_i) r + \beta = 0.$$

Discuss what happens in the following two cases:

(i) Flat direction: $\lambda = 0$.
Show that in this case

$$r^2 - (1 + \beta) r + \beta.$$

What are the roots and what does this imply about the motion in a perfectly flat direction?

(ii) Negative curvature: $\lambda < 0$.
Let

$$p(r) = r^2 - (1 + \beta - \eta\lambda) r + \beta.$$

Show that $p(1) = \eta\lambda < 0$.
Further argue that there must be a root satisfying $r > 1$. (*Hint*: Note that $p(r) \to +\infty$ as $r \to +\infty$).
What does this say about the stability of momentum close to a saddle point? Why might instability along a direction of negative curvature actually be useful during neural network optimization?